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Artificial Intelligence Assignment 2: Computer Science 311

1. Name and Purpose of the Chatbot

AI Assistant. Its purpose is to provide users with an intelligent, conversational interface to query specific product details, troubleshooting steps, and general information. By utilizing a Large Language Model (LLM) grounded in a specific product dataset, the chatbot reduces the need for manual searching through dense documentation while ensuring answers are accurate and context-aware.

2. NLP/LLM Methods Used

Method: Retrieval-Augmented Generation (RAG) using LangChain.

- **Justification:** Unlike standard LLMs that may "hallucinate" (invent facts), RAG ensures the chatbot only answers based on the provided Kaggle dataset.
- **Semantic Search:** By using **Vector Embeddings**, the bot understands the *meaning* behind a user's question rather than just matching keywords.
- **Prompt Engineering:** LangChain is used to "stuff" relevant document chunks into the LLM context window, forcing the model to act as a grounded product expert.

3. Dataset Information

- **Source:** Kaggle (e.g., [Name of the specific dataset you used]).
- **Features:** **Question** (String), **Answer/Description** (String), and **Category** (String).
- **Preprocessing:** 1. **Data Cleaning:** Used Pandas to remove **NaN** (null) values to prevent Pydantic validation errors. 2. **Type Casting:** Explicitly converted the text column to **String type** to ensure compatibility with embedding models. 3. **Text Chunking:** Applied a **RecursiveCharacterTextSplitter** to break long entries into **1,000-character chunks** with a **100-character overlap** to maintain contextual continuity.

5. Application Design and Implementation

- **Data Flow:** The system loads the CSV via Pandas, which is then processed into LangChain **Document** objects.
- **Storage:** These documents are converted into numerical vectors using OpenAI's **text-embedding-3-small** and stored in a **FAISS** (Facebook AI Similarity Search) vector database.

- **Retrieval:** When a user asks a question, the system searches the FAISS index for the most mathematically similar text chunks.
- **Generation:** The retrieved chunks and the user's question are sent to the `gpt-3.5-turbo` model via a **RetrievalQA** chain to generate a final, human-like response.